

TSM 098 | INDEPENDENT PRACTICE

Multiplying and dividing numerical fractions

While TSM 099 studies Topic 20. Review: Topic 19.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: multiplying fractions

1. Evaluate and simplify.

$$\frac{3}{5} \cdot \left(\frac{2}{10}\right)$$

2. Evaluate and simplify.

$$\frac{-6}{8} \cdot \left(\frac{3}{2}\right)$$

3. Evaluate and simplify.

$$\frac{7}{6} \cdot \left(\frac{-2}{2}\right)$$

4. Evaluate and simplify.

$$\frac{3}{5} \cdot \left(\frac{8}{6}\right)$$

5. Evaluate and simplify.

$$\frac{-8}{6} \cdot \left(\frac{4}{3}\right)$$

6. Evaluate and simplify.

$$\frac{6}{10} \cdot \left(\frac{6}{4}\right)$$

7. Evaluate and simplify.

$$\frac{7}{3} \cdot \left(\frac{-9}{10}\right)$$

8. Evaluate and simplify.

$$\frac{-8}{6} \cdot \left(\frac{2}{3}\right)$$

TSM 098 | INDEPENDENT PRACTICE

Multiplying and dividing numerical fractions

While TSM 099 studies Topic 20. Review: Topic 19.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: dividing fractions

9. Evaluate and simplify.

$$\frac{4}{3} \div \left(\frac{4}{2}\right)$$

10. Evaluate and simplify.

$$\frac{-2}{4} \div \left(\frac{5}{8}\right)$$

11. Evaluate and simplify.

$$\frac{5}{5} \div \left(\frac{-5}{3}\right)$$

12. Evaluate and simplify.

$$\frac{2}{3} \div \left(\frac{5}{8}\right)$$

13. Evaluate and simplify.

$$\frac{-4}{12} \div \left(\frac{4}{6}\right)$$

14. Evaluate and simplify.

$$\frac{4}{3} \div \left(\frac{1}{10}\right)$$

15. Evaluate and simplify.

$$\frac{5}{5} \div \left(\frac{-3}{8}\right)$$

16. Evaluate and simplify.

$$\frac{-6}{8} \div \left(\frac{8}{3}\right)$$

TSM 098 | INDEPENDENT PRACTICE

Multiplying and dividing numerical fractions

While TSM 099 studies Topic 20. Review: Topic 19.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: mixed multiplication and division

17. Evaluate and simplify.

$$\frac{5}{8} \cdot \left(\frac{7}{3}\right)$$

18. Evaluate and simplify.

$$\frac{-3}{6} \cdot \left(\frac{2}{8}\right)$$

19. Evaluate and simplify.

$$\frac{3}{12} \cdot \left(\frac{-7}{4}\right)$$

20. Evaluate and simplify.

$$\frac{4}{3} \cdot \left(\frac{1}{6}\right)$$

21. Evaluate and simplify.

$$\frac{4}{6} \div \left(\frac{3}{3}\right)$$

22. Evaluate and simplify.

$$\frac{-2}{3} \div \left(\frac{5}{2}\right)$$

23. Evaluate and simplify.

$$\frac{6}{10} \div \left(\frac{-7}{5}\right)$$

24. Evaluate and simplify.

$$\frac{9}{12} \div \left(\frac{1}{10}\right)$$

TSM 098 | INDEPENDENT PRACTICE

Multiplying and dividing numerical fractions

While TSM 099 studies Topic 20. Review: Topic 19.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Extension: longer expressions and reasoning

25. Evaluate and simplify. Show intermediate steps.

$$-1 \cdot \left(\frac{5}{8}\right) \div \left(\frac{1}{2}\right)$$

26. Evaluate and simplify. Show intermediate steps.

$$-\frac{5}{6} \cdot \left(-\frac{4}{5}\right) \div \left(\frac{1}{3}\right)$$

27. Evaluate and simplify. Show intermediate steps.

$$-\frac{5}{6} \cdot \left(\frac{3}{8}\right) \div \left(-\frac{3}{2}\right)$$

28. Evaluate and simplify. Show intermediate steps.

$$-1 \cdot \left(-\frac{4}{5}\right) \div \left(-\frac{3}{4}\right)$$

29. Evaluate and simplify. Show intermediate steps.

$$-\frac{3}{4} \cdot \left(\frac{1}{3}\right) \div \left(\frac{1}{2}\right)$$

30. Evaluate and simplify. Show intermediate steps.

$$\frac{4}{3} \cdot \left(\frac{2}{5}\right) \div \left(-\frac{3}{2}\right)$$

31. Explain why dividing by a fraction is different from multiplying by that fraction.

$$\frac{2}{3} \div \frac{4}{5}$$

32. Determine whether the expression is defined. Explain.

$$\frac{3}{7} \div 0$$